Module		Water Infrastructure: Channel Design
Topic	Sediment properties	

In order to understand sediment movement for the design of open channels, it is necessary to define and quantify the properties of sediments. Although you should be familiar with some of these terms from geotech courses, you will find that the parameters describe here are specifically relevant to sediment transport in a water medium.

Sediment size

Engineers often use sieve size as a measure of the size of sediment. Sieve size is the size of opening through which a particle will just pass. This is an arbitrary length which depends on the shape of the particle and may not tell us much about the behaviour of the particle in water. It is, however, very easily determined and used extensively for sands and gravels. Sieve size diameters for typical sediment size classes are defined in Table 1.

Table 1. Sieve size diameter for various sediment types

Sieve size Diameter (d)	*Sediment type*
---- < d < 0.004 mm	Clay
0.004 < d < 0.062 mm	Silt
0.062 < d < 2.0 mm	Sand
2.0 < d < 64 mm	Gravel
64 < d < ----	Cobbles and boulders

Sediment diameter size in hydraulic engineering is usually represented with a small "d" followed by a sub-script. Common uses of this format are described below:

d_p is the sieve opening which passes p% by weight. Thus d_p increases as p increases.
d_{50} is the median.

d_g refers to the geometric mean = $(d_{84.1} \times d_{15.9})^{1/2}$
(84.1% and 15.9% are one standard deviation from median)
d_{mm} = is just a diameter size of particles in millimetres

The geometric standard deviation is dimensionless and greater than or equal to 1.0 (uniform material).

σ_g is the geometric standard deviation = $(d_{84.1}/ d_{15.9})^{1/2}$

Quasi uniform material: $1.0 < \sigma_g < 1.3$
Non-uniform material: $1.3 < \sigma_g$

Some problems associated with σ_g are:

- long tails on the distribution
- bi-modal distributions: gravel bed rivers sometimes have sizes 2-8 mm underrepresented giving a frequency distribution with two peaks and a % finer distribution with a flatter section.

You may also encounter something called a Phi index scale:

$$\phi = -\log_2(d_{mm}) = -\frac{\log d_{mm}}{\log 2} \qquad [1]$$

d_{mm}	8	4	2	1	0.5	0.25	0.125	0.0625
ϕ	-3	-2	-1	0	1	2	3	4

Sediments in rivers and streams contain a range of particle sizes. Uniform sediments are rare and seldom occur outside the laboratory. Sieve analysis results on a percentage by weight finer than a sieve diameter and this can be plotted in a computer to obtain distributions. Typical values of sediment sizes found in rivers around the world and New Zealand are shown in Table 2.

Table 2. Typical values of particle sizes from various sources. (Hicks et al. 1994)

River	**location**	**$d_{50\ (mm)}$**	**σ_g**
Missouri river	Omaha	0.20	1.20
Colorado river	Taylor Ferry	0.32	1.44
Waikato river	Karapiro	6.49	3.73
Crest River	Skew Bridge	19.2	1.85
Waimakariri	SH1 Bridge	28.0	2.6
Flood gate creek	Kaikoura	170	5.40

Shape factor

The shape factor (S.F) is important in determining the behaviour of a particle in water. The shape factor can affect the entrainment of sediment (detachment of sediment from the bed) and the fall velocity (speed at which sediment will fall in a column of water). Transportation rates may also depend on the shape factor.

$$S.F. = \frac{c}{\sqrt{ab}} \qquad [2]$$

Where a, b, c are the largest, middle, and smallest axes of the grain of sediment. b approximates the sieve diameter. A sphere has a shape factor of 1.

Natural materials have shape factors ranging from $0.5 < S.F. < 0.9$
0.7 can be used for most river material although some material in Canterbury is flat so the shape factor is closer to 0.5.

Fall velocity

Fall velocity (ω) is the terminal velocity of an isolated particle in a fluid (water) at rest. It is an important parameter in many sediment transport calculations because it is a measure of the resistance offered to movement by the particle. Researchers have developed many ways of estimating fall velocities for different particles. An example of a simplified graph used to calculate fall velocity for sand sized material is shown in Figure 1.

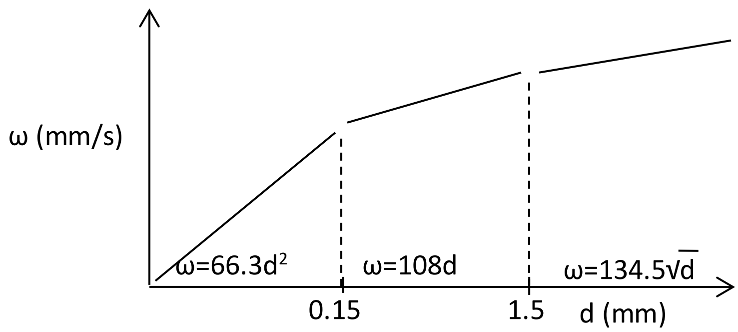

Figure 1. Simplified graph method of estimating fall velocity.

Since sediment shapes are not usually spherical, the concept of sedimentation diameter is often used. Sedimentation diameter is the diameter of a sphere of the same specific weight and fall velocity (terminal) as the given particle in the same fluid. Another simplification that is often made is that the fall velocity is calculated for discrete particles. This refers to settling of particles which fall independent of each other in a low-concentration solution. Using that assumption, fall velocity can be assumed to fall in turbulent free water as a response to the difference in the submerged weight of the particle and the drag of the fluid on the particle. At steady state, this can be described as follows:

$$\frac{\pi d^3}{6}(\gamma_s - \gamma) = C_D \frac{1}{2}\rho\omega^2 \frac{\pi d^2}{4} \qquad [3]$$

Where,

C_D is the coefficient of drag (can be obtained from different sources, C_D =1.2 is a typical value for spherical quartz sands).
S_s is the specific gravity of sand
g is gravity
d is the diameter of the particle

This can be simplified to:

$$\omega^2 = \frac{4}{3C_D}(S_s - 1)gd \qquad [4]$$

Stokes showed that the C_D for spheres depends on the Reynold's number:

$$C_D=24/R_e \qquad [5]$$

Where R_e is the Reynold's number given by:

$$R_e=\omega d/v \qquad [6]$$

Where v is the kinematic viscosity. Eqn. [6] is normally valid up to Re = 0.5 and the equations can be simplified to:

$$\omega = \frac{1}{18}\left[\frac{d^2 g}{\upsilon}(SG-1)\right] \qquad [7]$$

Parameters to remember
For normal temperature (20°C), use the following properties

Water:
$\gamma = 9810\ N/m^3$ (specific weight)
$\rho = 1000\ kg/m^3$ (density)
specific gravity: 1.00

Sand:
$\gamma_s = 2.65 \times 9810\ N/m^3$ (specific weight)
$\rho_s = 2650\ kg/m^3$ (density)

Specific gravity of sand (S_s): 2.65
$2.65 = \gamma_s / \gamma$

Sediments in Suspension

Concentration:

C = (dry weight of sediment in a volume of fluid) / (weight of volume of fluid)
= (mass of sediment) / (mass of fluid)
= (dry weight of sediment) / (weight of sample)

This is often used in practice and except for very high concentrations the error is negligible.

Concentration by volume:

C_v = (volume of sediment) / (volume of fluid)
= (mass of sediment) / (mass of fluid) * (γ / γ_s)

$$C_v = \frac{\gamma}{\gamma_s} C \quad \text{and} \quad C = \frac{\gamma_s}{\gamma} C_v$$

<u>Units:</u>

1.) concentrations are dimensionless as defined
2.) ppm (parts per million by dry weight)
1 ppm = (1 N of sediment) / (10^6 N of sample)
$C_{ppm} = 10^6$ C

3.) grams/litre - a mixed unit using mass and volume

C = (kg of sediment) / (kg of fluid) = (kg of sediment) / (litre of water)

= 1/1000 (grams of sediment)/(litres of water)

$C = 1/1000\ C_{g/l}$

4.) grams/m^3 will be 1000 times C g/l and thus 10^6C – same as ppm

Hence,

$C_{ppm} = C_{g/m}3 = 10^3 C_{g/l} = C_{mg/l} = 10^6 C$

Values in rivers range from 5 ppm – 5000 ppm (0.005 to 5 g/l)

For good fishing here in NZ, it is recommended that the water have ~ 20 ppm. At 80 ppm fish start to be stressed. Over 100 ppm for long periods can adversely affect fish!!

References

Hicks, D. M., and Fenwick, J. K. (1994). *Suspended sediment manual*, National Institute of Water and Atmospheric Research (Niwa), Christchurch, N.Z.

Hicks, D. M., Hill, J., and Shankar, U. 1996 "Variation of suspended sediment yields around New Zealand: the relative importance of rainfall and geology." *Erosion and Sediment Yield: Global and Regional Perspectives. Exeter Symposium*.